

Replicating the “美债因子看板 (US Rates Factor Desk)” website

1 Overview of the existing site

The website `us-treasury-bonds-monitor-luffa.vercel.app` is a single-page dashboard presenting a daily snapshot of the US Treasury market in Chinese and English. It is organised into tabbed sections: 摘要 (Summary), 收益率曲线 (Yield Curve), 成分分解 (Yield Decomposition), 因子计分卡 (Factor scorecard), 货币政策 (Monetary policy), 供给与拍卖 (Supply & auctions), 持仓与资金流 (Demand & positioning), 跨市场 (Cross-market), 事件与新闻 (Events & news) and 投资观点 (Investment opinions).

The page is **static**: in the `view-source` the entire data used by the charts and tables is defined in a single `DATA` object. The object contains the “as-of” date, yield curve arrays (`today` , `w1` and `m1`), Fed-funds path probabilities, news headlines and a nested list of factor groups with weights and scores. The yield curve arrays for 1 M, 3 M, ... 30 Y and the corresponding daily, weekly and monthly changes are encoded as arrays in the data object ¹ . The policy-path probabilities are defined for upcoming FOMC meetings (`hike` , `hold` , `cut`) ² , and news items are included as simple strings ³ . Six factor groups (Monetary policy, Macro fundamentals, Supply & technicals, Demand & positioning, Relative value, Sentiment & liquidity) contain individual factors with descriptive tags, current values and a discrete score between -2 and +2 ⁴ ⁵ .

The front-end code renders charts and tables by drawing SVGs directly in JavaScript. There is **no network request in the page**, which implies that the author updates the `DATA` object manually or via a build script each day. Our replication effort will therefore need to reproduce the UI as well as build a **back-end data pipeline** so that the dashboard can be updated automatically from reliable data sources.

2 Objectives

1. **Functional replication**: build a web application that mirrors the structure and functionality of the existing dashboard: hero tiles (2 Y, 5 Y, 10 Y, 30 Y yields and slope metrics), yield-curve charts and tables, yield-decomposition visualisation, factor scorecard, policy path chart, sections on supply/auctions, demand/flows, cross-market comparisons, events & news and investment viewpoints.
2. **Data automation**: design a data layer that automatically gathers up-to-date market data (yields, breakeven inflation, term premium, Fed-funds futures, auction results, CFTC positioning, TIC holdings, etc.) from authoritative sources rather than manually entering values.
3. **Scalability and maintainability**: choose modern frameworks (e.g., Next.js + React) for an easily maintainable front-end, and implement scheduled ETL scripts to update the data repository daily.
4. **Internationalisation**: support both Chinese and English text as in the original site.
5. **Deployment**: host the application on a cloud platform (e.g., Vercel) with CI/CD and monitoring.

3 Feature analysis and data requirements

3.1 Summary (Hero section)

- **Yields & slopes:** Show the latest yields for key tenors (2 Y, 5 Y, 10 Y, 30 Y) and the slopes (2s10s and 5s30s). The existing site stores the latest yields and daily changes in arrays ¹ and computes slopes on the fly ⁶.
- **Market regime:** A text description summarises the current macro regime (e.g., “inflation shock / policy standoff”). This will need to be generated manually or by an analyst.
- **Duration & curve stance:** An automated inference of whether to be long or short duration (“看空久期” / “看多久期”) and whether to favour curve steepening or flattening, based on the aggregate factor score. This mapping is hard-coded in the original script and can be replicated ⁷.

Required data: Daily Treasury yield curve, previous day/week/month yields, slope calculations.

3.2 Yield curve

- **Curve chart and table:** Plot the par yield curve for 11 maturities (1M to 30Y) and overlay it with the curves from one week ago and one month ago. The site also displays a table of yields and changes.
- **Metrics:** compute slopes (2s10s, 5s30s, 3M–10Y) and a butterfly (“2s5s10s”) for curvature ⁸.

Required data:

- **Daily Treasury Par Yield Curve Rates:** The U.S. Treasury publishes daily par yield curve rates. Historical data can be downloaded as CSV files (e.g., `par-yield-curve-rates-2020-2023.csv`) with columns for dates and maturities ⁹. An API is not officially provided, so a daily script should download the latest CSV and parse the required tenors.
- **Constant-maturity yields (FRED):** Alternatively, FRED provides series such as GS1M (1-Month), GS1 (1-Year), GS2 (2-Year), GS5, GS7, GS10, GS20 and GS30. The FRED API (e.g., `https://api.stlouisfed.org/fred/series/observations?series_id=GS10&api_key=...&file_type=json`) returns daily observations. These series can be used to compute weekly and monthly changes.
- **Short-term real yield & breakeven inflation:** FRED series `DFII10` (10-year TIPS real yield) and `T10YIE` (10-year breakeven inflation) allow calculation of expected inflation (real + inflation = nominal).

3.3 Yield decomposition (Clarida four-factor)

The site decomposes the 10-year nominal yield into four components: **expected real short rate** ($E[r_{\text{short}}]$), **expected short-term inflation** ($E[\pi_{\text{short}}]$), **real term premium**, and **inflation risk premium** [192646928754690 + L1190-L166]. The decomposition is described qualitatively in the data object. To replicate this quantitatively:

- **Expected real short rate:** Approximate using the current Effective Federal Funds Rate (FRED series `FEDFUNDS`) minus short-term inflation expectations (e.g., 1-year breakeven inflation or 1-year inflation swap).
- **Expected short-term inflation:** Use survey-based measures or breakeven inflation series (FRED `T5YIFR` for 5-year forward inflation expectation).

- **Real term premium:** Use the **Adrian-Crump-Moench (ACM)** term premium estimates from the New York Fed. The bank provides term premium estimates for maturities from 1 Y to 10 Y. The dataset is downloadable via their website (CSV).
- **Inflation risk premium:** Derive as the difference between the nominal term premium and the real term premium or use estimates from academic papers.

3.4 Policy path probabilities

The “Fed path” chart shows the probabilities of rate hikes, holds or cuts for the next five FOMC meetings, as percentages ². These probabilities are implicitly derived from Fed-funds futures prices.

Required data & method:

- Download daily prices of Fed-funds futures contracts (e.g., from CME Group data or via a data vendor).
- Convert futures prices into implied target rates and compute the probability of each outcome assuming a binary distribution around the nearest Federal Reserve target range.
- Alternatively, use the CME FedWatch tool’s data if an API is available.

3.5 Factor scorecard

Six groups of factors summarise monetary policy, macro fundamentals, supply & technicals, demand & positioning, relative value and sentiment & liquidity. Each factor has a descriptive tag, a qualitative value and a score from –2 (bearish) to +2 (bullish). The aggregated weighted score drives the duration and curve stances.

Required data sources:

Factor group	Examples in the current data	Suggested data sources
Monetary policy	Fed target rate, implied policy path, Fed leadership attitude, balance-sheet stance ⁴	FOMC statements & minutes (Fed website); Fed Funds futures data; Federal Reserve board press releases; for balance-sheet, track Treasury securities held outright (New York Fed SOMA data).
Macro fundamentals	CPI, PPI, labour market, GDP growth ¹⁰	U.S. Bureau of Labor Statistics (CPI CPIAUCSL , PPI PPIACO via FRED); Bureau of Economic Analysis (GDP GDPC1); employment (UNRATE).
Supply & technicals	Recent auctions (coupon rate, bid-to-cover), issuance pace (QRA), TGA level ⁵	U.S. Treasury auction results (TreasuryDirect “Marketable Securities Auctions” API or CSV); Treasury Quarterly Refunding statements; Treasury General Account (TGA) balance from FRED series WTREGEN .
Demand & positioning	Leveraged funds net positions, TIC holdings, primary dealer inventory ¹¹	CFTC Commitments of Traders reports (weekly “disaggregated” data for Treasury futures); Treasury International Capital (TIC) monthly holdings (CSV/Excel from treasury.gov); New York Fed “Primary Dealer Statistics” weekly.

Factor group	Examples in the current data	Suggested data sources
Relative value	Term premium (ACM), real yield vs breakeven, swap spreads ¹²	ACM term premium (New York Fed CSV); real yield (FRED <code>DFII10</code>); breakeven inflation (FRED <code>T10YIE</code>); swap spreads from market data provider or published daily on some financial platforms.
Sentiment & liquidity	MOVE index, market liquidity, on-the-run/off-the-run spread ¹³	MOVE index values (ICE BofA; accessible via data vendors); bid-ask spreads and market depth from broker feeds; for on-the-run vs off-the-run yield differences, compute from yield curve data.

3.6 Supply, auctions and funding

A dedicated section should track upcoming and recent Treasury auctions (coupon, notes and bonds). Data can be scraped from TreasuryDirect's auction announcement and result pages or via the Treasury API (e.g., <https://www.treasurydirect.gov/auctions/announcements.csv>). The plan should also capture **government cash balances** (TGA), the **debt ceiling situation** and **quarterly refunding** details.

3.7 Demand & positioning (holdings & flows)

Track the latest CFTC leveraged funds net positioning in Treasury futures, monthly TIC data summarising foreign holdings of U.S. Treasuries (available in CSV) and the weekly net positions of primary dealers (New York Fed). Present these metrics in a small table or bar chart with directional signals.

3.8 Cross-market and events/news

- **Cross-market:** Compare the U.S. Treasury yield curve against other markets (e.g., German bund yields, Japanese Government Bond yields, inflation-linked bonds or swap curves). Data can be sourced from the European Central Bank's SDW (for German yields) or the Bank of Japan's yield curve data via their open-data APIs.
- **Events & news:** Provide a timeline of important economic releases (CPI, PPI, FOMC dates, employment reports) and major news headlines. Use RSS feeds from CNBC/Bloomberg or manually enter curated headlines.

3.9 Investment viewpoints

This section summarises the analyst's view on the bond market and recommended trading strategies (e.g., duration bias, curve trades, relative value trades). It may not be derived automatically, so content management functionality should allow manual updates.

3.10 Data update frequencies and sources

Because the dashboard is meant to refresh automatically every business day, it is critical to choose data sources with clear update schedules. The table below summarises the key datasets needed for the US Rates Factor Desk and shows when each series is updated and where to retrieve it. Only those sources with daily updates (or faster) should be polled each business day; weekly or monthly series can be updated on their release dates. Citations are provided from official websites to justify the update cadence.

Dataset / indicator	Update frequency / timing	Evidence	Notes
Treasury par yield curve (1M–30Y)	Daily on business days ; the Federal Reserve Bank of New York provides closing market bid prices at ~3:30 p.m. each business day for the most recently auctioned securities ¹⁴ .	U.S. Treasury’ s Interest Rate Statistics page states that the par yield curve is derived from closing market bid prices obtained from the Federal Reserve Bank of New York each business day ¹⁴ .	Download daily CSVs from Treasury.gov (no API).
Constant-maturity yields (H.15)	Daily (Mon-Fri) ; release posted at 4:15 p.m. ET, except holidays ¹⁵ .	The Federal Reserve Board’ s H.15 release explicitly notes that the data are posted daily Monday-Friday at 4:15 p.m. ET ¹⁵ .	Use FRED API or Fed Board CSV to fetch GS1M, GS1, GS2, GS5, GS7, GS10, GS20 and GS30.
Term premium (Adrian–Crump–Moench)	Daily and monthly series available ¹⁶ .	The New York Fed notes that the ACM term-premium estimates are provided at daily and monthly frequencies ¹⁶ .	Fetch CSV from New York Fed website and update daily.
Effective federal funds rate (FRED series <code>FEDFUNDS</code>)	Daily	FRED series metadata (not cited here due to site restrictions) show the series is “Daily, percent, not seasonally adjusted.”	Use FRED API.
H.4.1 factors affecting reserve balances (Fed balance-sheet)	Weekly ; released each Thursday at 4:30 p.m. ET ¹⁷ .	The Federal Reserve Board states that the H.4.1 statistical release is normally issued each Thursday at 4:30 p.m. ET ¹⁷ .	Update weekly; incorporate totals of Treasury securities held outright.
Primary dealer positions & transactions	Weekly ¹⁸ .	The New York Fed explains that primary dealers report positions and trading activity weekly, and the data are released weekly ¹⁸ .	Use the Primary Dealer Statistics Excel file published weekly.

Dataset / indicator	Update frequency / timing	Evidence	Notes
CFTC Commitment of Traders (Treasury futures positioning)	Weekly ; released Fridays at 3:30 p.m. ET, covering data through the preceding Tuesday ¹⁹ .	The CFTC release schedule notes that COT reports are published at 3:30 p.m. ET each Friday (or on Monday when holidays delay the release), reflecting positions from Tuesday of that week ¹⁹ .	Download the Disaggregated Commitments of Traders report and parse non-commercial leveraged fund positions.
Treasury International Capital (TIC) flows	Monthly ; release dates are typically six weeks after month-end. For example, the March 2026 press release notes that the next release (for April data) will be on 18 June 2026 ²⁰ .	Treasury’ s TIC press release schedule shows that monthly TIC data are published with a six-week lag ²⁰ .	Update monthly when data are released.
Treasury auction results	Real-time on auction days ; TreasuryDirect updates the page as soon as results are available.	TreasuryDirect’ s “Today’ s Auction Results” page states that results are updated in real time when auctions occur.	Scrape or subscribe to the auction results feed on each auction day.
S&P 500 index level	Daily close ; FRED documentation for the S&P 500 series notes that observations represent the daily index value at market close (U.S. stock market typically closes at 4 p.m. ET) ²¹ .	The FRED PDF describing the S&P 500 index states that the data represent the daily index value at market close, normally 4 p.m. ET ²¹ .	Use FRED API or another provider for daily closing values.
VIX volatility index	Updated throughout trading day; available during extended trading hours ²² .	A Cboe white paper explains that the VIX index is disseminated in extended trading hours (2:00–8:15 a.m. CT) and during regular trading hours (8:30 a.m.–3:15 p.m. CT), with a smoothing algorithm applied ²² .	Treat the VIX as a daily (intraday) series; fetch the latest close each trading day.

Dataset / indicator	Update frequency / timing	Evidence	Notes
Crude oil (Brent & WTI spot prices)	Daily closing prices ; the U.S. Energy Information Administration (EIA) calculates weekly/monthly averages from daily closing spot prices ²³ .	The EIA' s spot-price table notes that weekly, monthly and annual prices are derived from daily closing spot prices ²³ .	Download the daily spot price table from EIA each business day.
Gold price (London fix)	Twice daily ; gold prices are fixed at 10:30 a.m. and 3:00 p.m. London time ²⁴ .	Moody' s Analytics explains that the LBMA sets gold prices twice a day (10:30 a.m. and 3:00 p.m. London time) ²⁴ .	Use the PM fix (3 p.m. London) as the daily close value.
MOVE index (Treasury volatility)	Daily (not cited due to limited access).	The MOVE index from ICE BofA is widely reported as a daily closing index.	Use data vendor or FRED if available.
Dollar index (DXY / ICE USDX)	Continuous trading with daily settlement ; the ICE USDX futures contract specifies a daily settlement based on the volume-weighted average price in the closing session from 14:59–15:00 ET ²⁵ .	The ICE USDX contract specifications note that daily settlement occurs at the volume-weighted average price in the closing session ²⁶ , implying a daily price.	Obtain daily closing values from a market data provider.
Euro-area government bond yield curves (Bund proxy)	Daily ; the ECB estimates euro-area government bond yield curves and publishes the par, spot and forward curves on a daily basis at noon ²⁷ ²⁸ .	The ECB' s yield-curve methodology page states that the euro-area yield curves are published on a daily basis at noon, with daily data available from 6 September 2004 onward ²⁷ ²⁸ .	Download the daily yield curve CSV from the ECB Data Portal; use the 10-year par yield as a proxy for the German Bund yield.
UK Gilt yield curves	Daily ; the Bank of England aims to publish its yield curves by noon of the following business day ²⁹ .	The Bank of England' s yield-curves FAQ explains that the site publishes yield curves by noon of the following business day ²⁹ .	Use the Bank of England' s daily yield-curve spreadsheet; yields are derived from fitted curves using Bloomberg/ Refinitiv data.

Dataset / indicator	Update frequency / timing	Evidence	Notes
Japanese government bond (JGB) yields	Daily (citation not accessible); the Bank of Japan and Japan Bond Trading Co. publish daily closing yields for benchmark JGBs.	Due to website restrictions, we could not access an official citation; however, JGB yields are widely reported as daily closing values.	Obtain daily JGB yields via data vendors or FRED series (e.g., <code>IRLTLT01JPM156N</code>).
ICE BofA corporate bond indices (Investment-grade & High-yield)	Daily ; index values and analytics are provided in a daily index level file (e.g., <code>Index_Level_YYYYMMDD.csv</code>) ³⁰ , and the methodology lists a daily total return percentage ³¹ .	The STOXX/ICE Fixed Income Indices methodology guide notes that index values are displayed in the daily index level file and that bond analytics (e.g., effective yield, spreads, duration) are included in daily constituent files ³⁰ ³² .	Use the ICE Data Indices feed or data vendor (Bloomberg, Refinitiv) to retrieve daily effective yields and option-adjusted spreads for investment-grade and high-yield corporate indices.
UK gilt reference prices and yields	Daily ; end-of-day reference prices for gilts, STRIPS and Treasury bills are produced by FTSE-Tradeweb and made available at noon of the day following initial publication ³³ .	The UK Debt Management Office explains that end-of-day reference prices for gilts are administered by FTSE-Tradeweb and made available to the public from 12:00 noon on the day after publication ³³ .	Use these reference prices to compute daily gilt yields and spreads.

The ETL layer described in § 4.1 should schedule separate jobs based on these update frequencies: daily jobs for high-frequency series (yield curves, term premium, market indices, oil/gold prices), weekly jobs for H.4.1, primary dealer statistics and COT data, and monthly jobs for TIC and other low-frequency indicators.

4 System architecture

4.1 High-level design

1. **Data ingestion layer:** A set of scheduled scripts (Python) fetches data from external sources daily. They download Treasury yield CSVs, query FRED API for macro data, scrape auction results, fetch CFTC and TIC reports and compute derived measures (slopes, term premium, breakeven inflation, scoring metrics). The results are stored in a structured format (e.g., JSON file or PostgreSQL database).
2. **Backend API:** A lightweight REST API (e.g., built with FastAPI or Next.js API routes) exposes aggregated data to the front-end. It reads from the database or JSON and returns endpoints such

as `/api/curve`, `/api/fed_path`, `/api/scorecard`, `/api/auctions` and `/api/news`. This API can also provide endpoint parameters for date ranges or exporting CSV.

3. **Front-end:** A React/Next.js application replicates the original dashboard's aesthetic. It fetches data from the backend API at build time or client-side and renders charts using a charting library (e.g., D3.js or ECharts). Components such as hero tiles, yield-curve chart, decomposition bars, factor scorecard with collapsible groups, and tables will be created. Internationalisation (i18n) support is implemented via libraries like `next-i18next`.
4. **Content management:** A simple admin interface (or editable Markdown files) allows analysts to update narrative text (market regime, investment viewpoints, notes) and factor weightings.
5. **Deployment:** The application is deployed to Vercel (for Next.js) or a similar platform with environment variables for API keys. Data ingestion jobs can run on GitHub Actions (daily cron) or on a serverless task scheduler.

4.2 Technology stack

- **Data processing:** Python 3 with `pandas`, `requests`, `schedule/cron`, `fredapi` (FRED API wrapper), `beautifulsoup4` / `lxml` for scraping, and Jupyter notebooks for development.
- **Database/storage:** Small-scale use can rely on a JSON file checked into the repository. For larger scale or multi-user use, adopt SQLite or PostgreSQL.
- **Back-end:** Node.js with Express or FastAPI in Python. Next.js API routes may suffice for serverless deployments.
- **Front-end:** Next.js/React, Tailwind CSS or custom CSS modules (matching the CSS variables found in the original site). For charts, prefer **ECharts** or **D3.js** for full customisability; replicating the hand-drawn SVG used in the original site is also possible.
- **Internationalisation:** `next-i18next`, `react-intl` or `vue-i18n` (if using Vue instead of React).
- **Version control & CI:** Git with GitHub; GitHub Actions for linting, testing and scheduled data updates.
- **Hosting:** Vercel for the front-end and serverless back-end. Alternatively use AWS Amplify or Netlify.

5 Implementation plan and timeline

Phase	Duration	Tasks	Deliverables
1 – Requirements & design	1 week	▸ Interview stakeholders to confirm scope and desired features. ▸ Produce wire-frames replicating each section of the dashboard. ▸ Identify mandatory data feeds and confirm API availability and licensing.	Requirements document and wire-frames.

Phase	Duration	Tasks	Deliverables
2 – Data pipeline development	2 weeks	▸ Develop Python scripts to download and parse daily Treasury yield curve CSVs ⁹. ▸ Use FRED API to fetch daily series (yields, CPI, PPI, employment, BEI, real yields). ▸ Scrape Treasury auction announcements/results, TIC data and CFTC commitments. ▸ Compute derived metrics: slope (2s10s, 5s30s, 3M–10Y), curvature, real/nominal decomposition, ACM term premium, breakeven inflation. ▸ Create a scoring engine that assigns –2 to +2 scores based on thresholds or z-scores. ▸ Store results in structured JSON/database.	Automated ETL scripts; Data schema; Daily dataset updates.
3 – Back-end API	1 week	▸ Build RESTful endpoints to serve aggregated data (curve, decomposition, scorecard, policy path, auctions, positioning, cross-market). ▸ Include query parameters (e.g., date ranges). ▸ Implement caching.	API running locally with documentation.
4 – Front-end development	2 weeks	▸ Set up Next.js project and global styles replicating the original colour palette. ▸ Build modular React components for hero tiles, yield curve chart/table, decomposition bars, scorecard accordion, Fed path probabilities, supply/demand sections and news timeline. ▸ Implement i18n. ▸ Connect components to API endpoints. ▸ Ensure responsiveness across devices.	Fully functional front-end prototype.
5 – Integration & testing	1 week	▸ Connect the front-end to the live back-end and ensure data flows correctly. ▸ Write unit tests for data processing functions and end-to-end tests for the UI. ▸ Validate the computed metrics against trusted sources. ▸ Perform performance optimisation and accessibility testing.	Completed application ready for deployment.
6 – Deployment & documentation	1 week	▸ Set up Vercel project; configure environment variables and secrets for API keys. ▸ Configure GitHub Actions for automatic deployment and scheduled ETL jobs. ▸ Write user guide and developer documentation explaining how to add new factors or update text. ▸ Plan monitoring and alerts for data update failures.	Live website and documentation.

Total estimated duration: ~6 weeks. Adjustments may be needed depending on data-feed complexity and team size.

6 Key considerations and challenges

1. **Data availability & licensing:** Official Treasury yield data is open, but real-time term premium estimates or Fed-funds futures data may require paid feeds. Identify open-source alternatives (e.g., FRED series for constant-maturity yields, New York Fed’s public term-premium CSV) and clarify licensing.
2. **ETL reliability:** Many data sources publish at different times (Treasury yields after 3 pm ET, CFTC reports weekly). The update schedule must account for these timings. Implement retries and error logging.
3. **Subjective scores:** The factor scorecard involves qualitative judgement. The scoring engine should be parameterised so that analysts can adjust thresholds. It might be useful to provide a manual override in the admin panel.
4. **Internationalisation:** Chinese translations need careful proofreading by native speakers.
5. **Extendability:** Keep the architecture modular to allow additional asset classes (e.g., FX, commodities) to be added later.

7 Conclusion

The existing “美债因子看板” is a visually elegant but static dashboard. By separating the presentation layer from the data layer and designing an automated data pipeline, we can build a maintainable and extensible platform that provides timely insights into the US Treasury market. Leveraging open government datasets (Treasury yield curve CSVs ⁹, FRED series, TIC and CFTC reports) and implementing a modular API/React architecture will allow the dashboard to update automatically while preserving the original design aesthetics. The detailed plan above outlines the required data sources, processing steps, technology choices and development timeline, providing a roadmap for successful implementation.

1 2 3 4 5 6 7 8 10 11 12 13 美债因子看板 • US Rates Factor Desk

<https://us-treasury-bonds-monitor-luffa.vercel.app/>

⁹ home.treasury.gov

<https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rate-archives/par-yield-curve-rates-2020-2023.csv>

¹⁴ Interest Rate Statistics | U.S. Department of the Treasury

<https://home.treasury.gov/policy-issues/financing-the-government/interest-rate-statistics>

¹⁵ Federal Reserve Board - H.15 - Selected Interest Rates (Daily) - May 19, 2026

<https://www.federalreserve.gov/releases/h15/>

¹⁶ Treasury Term Premia - FEDERAL RESERVE BANK of NEW YORK

https://www.newyorkfed.org/research/data_indicators/term-premia-tabs

¹⁷ Federal Reserve Balance Sheet: Factors Affecting Reserve Balances - H.4.1 - Release Dates

<https://www.federalreserve.gov/releases/h41/>

¹⁸ Primary Dealers - Federal Reserve Bank of New York - FEDERAL RESERVE BANK of NEW YORK

<https://www.newyorkfed.org/markets/primarydealers>

¹⁹ Release Schedule | CFTC

<https://www.cftc.gov/MarketReports/CommitmentsofTraders/ReleaseSchedule/index.htm>

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22 419_VIX.pdf

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23 Spot Prices for Crude Oil and Petroleum Products

https://www.eia.gov/dnav/pet/pet_pri_spt_s1_d.htm

24 United States Gold Price: London PM Fix | Moody's Analytics

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25 26 ICE_USDX_Brochure.pdf

https://www.ice.com/publicdocs/ICE_USDX_Brochure.pdf

27 28 Yield curves | ECB Data Portal

<https://data.ecb.europa.eu/methodology/yield-curves>

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30 31 32 INDEX GUIDE

https://www.stoxx.com/document/Indices/Common/Indexguide/stoxx_ice_fixed_income_index_methodology_guide.pdf

33 Historical Prices and Yields

<https://www.dmo.gov.uk/data/gilt-market/historical-prices-and-yields/>